Double brooms refute Graffiti.pc Conjecture 340 on total domination in trees

Jonas J. Gebendorfer

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Abstract

Conjecture 340 of DeLaviña's collection *Written on the Wall II* asserts that every tree T of order $n > 2$ satisfies $\gamma_t(T) \le c(\langle N(L) \cup L\rangle) + \mathrm{mode}_{\min}(T)\,\gamma(T)$, where L is the set of leaves, c counts components, and $\mathrm{mode}_{\min}(T)$ is the smallest most frequent vertex degree. We show that the conjecture is false. The double broom obtained from a path on twelve vertices by attaching a leaves to one end and b leaves to the other, with $a, b \ge 2$ and $a + b \ge 10$, has total domination number 8 while the right-hand side equals 7. This gives counterexamples of every order $n \ge 22$. A one-ended broom family shows that the additive deficit is unbounded, and that no inequality $\gamma_t \le c(\langle N(L) \cup L\rangle) + \alpha\gamma + C$ with a constant $\alpha < 3/2$ can hold even among trees whose smallest mode is one, whereas $\alpha = 2$ is valid. All exact values are proved by explicit sets and disjoint-neighbourhood packings. An exhaustive census of all 9 114 283 trees of orders 3 through 22 shows that 22 is the smallest order of a counterexample and that the four double brooms of that order are the only ones. A standard-library supplement reproduces every finite statement.

Keywords. total domination; domination; trees; degree mode; Graffiti.pc; counterexample.
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1 Introduction

Graffiti.pc is DeLaviña's conjecture-generating program, a variant of Fajtlowicz's Graffiti [4]. Its conjectures are collected in *Written on the Wall II* (WOWII) [5]. A block of entries dated 18 February 2009 concerns the total domination number of trees; several of them are discussed in [6], and many have since been proved or refuted. Entry 340 is an upper bound. It combines the ordinary domination number with two simple structural quantities: the number of components of the subgraph induced by the leaves and their neighbours, and the smallest mode of the degree sequence.

The general inequality $\gamma_t(G) \le 2\gamma(G)$ for graphs without isolated vertices (Lemma 3.2) already gives an upper bound of the same shape with coefficient two. Conjecture 340 asks for much more when leaves are the most frequent degree: the coefficient of γ then drops to one. We show that this is false. The obstruction is elementary. A long path has γ_t/γ close to $3/2$, and a single bundle of leaves at each end of the path makes degree one the mode without creating any further leaf–support component.

Results. Theorem 4.3 gives counterexamples of every order $n \ge 22$, with fully hand-checkable certificates. Theorem 5.3 gives a second, one-ended family of counterexamples of every order $n \ge 23$. Theorem 5.5 shows that the additive deficit is unbounded. Theorem 5.6 shows that,

among trees whose smallest mode is one, the coefficient of γ cannot be lowered below $3/2$ even with an additive constant, while the coefficient 2 is valid for all trees. Proposition 6.1 records an exhaustive computation: no tree of order at most 21 is a counterexample, and exactly four trees of order 22 are, namely the double brooms of Theorem 4.3 with $a + b = 10$. Only Proposition 6.1 depends on computation.

2 Definitions and the source statement

All graphs are finite, simple and undirected. For a vertex v, $N(v)$ is its open neighbourhood and $N[v] = N(v) \cup \{v\}$ its closed neighbourhood; for a set X, $N(X) = \bigcup_{x \in X} N(x)$. The subgraph induced by X is $\langle X \rangle$, and $c(H)$ is the number of connected components of a graph H. A *leaf* is a vertex of degree one, and a *support vertex* is a vertex adjacent to a leaf. We write $L = L(T)$ and $S = S(T)$ for the sets of leaves and support vertices of a tree T.

A set $D \subseteq V(G)$ is *dominating* if $D \cap N[v] \neq \emptyset$ for every vertex v, and *total dominating* if $D \cap N(v) \neq \emptyset$ for every vertex v, including the vertices of D itself. Their minimum sizes are the domination number $\gamma(G)$ and the total domination number $\gamma_t(G)$; the latter exists when G has no isolated vertex [3, 9]. The *mode* of the degree sequence is a degree of maximum frequency, and $\mathrm{mode}_{\min}(G)$ is the smallest such degree.

The conjecture. The archived WOWII list [5] contains the following entry, with status marker O (open):

> **340.** *If T is a tree on $n > 2$ vertices, then $\gamma_t \leq$ number of components of $\langle N(L) \cup L \rangle +$ $\mathrm{mode}_{\min}(T) \cdot \gamma(T)$, where L is the set of leaves of T.* (dated Feb. 18, 2009)

The entry links WOWII definitions 94 (total domination number), 100 (induced subgraph), 45 (mode; "in case there is more than one mode, modemin is the smallest mode") and 49 (domination number); definition 46 defines $N(S)$ as the set of vertices adjacent to at least one vertex of S. In symbols, the conjecture states

$$\gamma_t(T) \;\leq\; c(\langle N(L) \cup L \rangle) + \mathrm{mode}_{\min}(T)\, \gamma(T) \qquad \text{for every tree } T \text{ of order } n > 2. \tag{1}$$

We write $R(T)$ for the right-hand side of (1).

Lemma 2.1. *If T is a tree of order $n > 2$, then $N(L) = S$ and $N(L) \cup L = S \cup L$. The same set arises if $N(L)$ is read as the closed neighbourhood $N[L]$.*

Proof. A vertex lies in $N(L)$ exactly when it is adjacent to a leaf, that is, when it is a support vertex; so $N(L) = S$. Moreover $N[L] = N(L) \cup L$. (Since two adjacent leaves would form a component K_2, which a tree of order $n > 2$ cannot have, S and L are even disjoint.) □

Thus the statement has no typographic ambiguity that affects the constructions below.

3 Three elementary lower and upper bounds

Lemma 3.1. *Let G be a graph and $v_1, \ldots, v_r$ distinct vertices.*

1. *If $N(v_1), \ldots, N(v_r)$ are pairwise disjoint and G has no isolated vertex, then $\gamma_t(G) \geq r$.*
2. *If $N[v_1], \ldots, N[v_r]$ are pairwise disjoint, then $\gamma(G) \geq r$.*

3. If $P \subseteq V(G)$ and every vertex of G has at most two neighbours in P, then $\gamma_t(G) \geq |P|/2$.

Proof. (1) A total dominating set meets each $N(v_i)$, and these sets are disjoint. (2) is the same argument for closed neighbourhoods. (3) If D is total dominating, every vertex of P has a neighbour in D, so $|P| \leq \sum_{x \in D} |N(x) \cap P| \leq 2|D|$. □

Lemma 3.2 (folklore)**.** *If G has no isolated vertex, then $\gamma_t(G) \leq 2\gamma(G)$.*

Proof. Let D be a minimum dominating set and choose for each $x \in D$ one neighbour x'. Every vertex has a neighbour in $D \cup \{x' : x \in D\}$: a vertex outside D has one in D, and each $x \in D$ is adjacent to x'. This set has at most $2\gamma(G)$ vertices. □

4 Double brooms

Definition 4.1. For integers $a, b \geq 2$, the *double broom* $D(a, b)$ consists of a path $w_1 w_2 \cdots w_{12}$, a further vertices $\ell_1, \ldots, \ell_a$ adjacent only to w_1, and b further vertices $\ell'_1, \ldots, \ell'_b$ adjacent only to w_{12}. It is a tree of order $12 + a + b$.

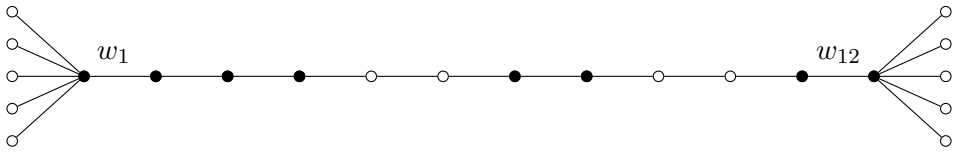


Figure 1: The double broom $D(5, 5)$ of order 22. The filled vertices form a minimum total dominating set of size 8, whereas $c(\langle N(L) \cup L \rangle) + \mathrm{mode}_{\min} \cdot \gamma = 2 + 1 \cdot 5 = 7$.

Lemma 4.2. *If $a, b \geq 2$ and $a + b \geq 10$, then $c(\langle N(L) \cup L \rangle) = 2$ and $\mathrm{mode}_{\min}(D(a, b)) = 1$; hence $R(D(a, b)) = 2 + \gamma(D(a, b))$.*

Proof. The leaves are the ℓ_i and the ℓ'_j, and the support vertices are w_1 and w_{12}. Thus $N(L) \cup L$ is the union of the vertex sets of the stars centred at w_1 and at w_{12}. No edge joins them, since w_1 and w_{12} are not adjacent and leaves have no further neighbours. Hence there are exactly two components. The degree sequence consists of $a + b$ ones, ten twos (the vertices $w_2, \ldots, w_{11}$), and the values $a + 1, b + 1 \geq 3$ attained by w_1 and w_{12}; every degree class other than those of 1 and 2 has at most two members. Because $a + b \geq 10$, degree 1 has maximum frequency, and it is the smallest degree present. Hence $\mathrm{mode}_{\min} = 1$. □

Theorem 4.3. *If $a, b \geq 2$ and $a + b \geq 10$, then*

$$\gamma_t(D(a, b)) = 8, \qquad \gamma(D(a, b)) = 5, \qquad R(D(a, b)) = 7.$$

In particular Conjecture 340 is false, and for every $n \geq 22$ there is a counterexample of order n.

Proof. Upper bound for γ_t. Let $X = \{w_1, w_2, w_3, w_4, w_7, w_8, w_{11}, w_{12}\}$. Every ℓ_i is adjacent to $w_1 \in X$ and every ℓ'_j to $w_{12} \in X$. The path vertices have the following neighbours in X:

v	w_1	w_2	w_3	w_4	w_5	w_6	w_7	w_8	w_9	w_{10}	w_{11}	w_{12}
neighbour in X	w_2	w_1	w_2	w_3	w_4	w_7	w_8	w_7	w_8	w_{11}	w_{12}	w_{11}

Hence X is total dominating and $\gamma_t \leq 8$.

Lower bound for γ_t. The eight vertices $\ell_1, w_1, w_4, w_5, w_8, w_9, w_{12}, \ell'_1$ have open neighbourhoods

$$\{w_1\},\ \{w_2, \ell_1, \dots, \ell_a\},\ \{w_3, w_5\},\ \{w_4, w_6\},\ \{w_7, w_9\},\ \{w_8, w_{10}\},\ \{w_{11}, \ell'_1, \dots, \ell'_b\},\ \{w_{12}\},$$

which are pairwise disjoint. By Lemma 3.1(1), $\gamma_t \geq 8$.

Domination. The set $\{w_1, w_4, w_7, w_{10}, w_{12}\}$ dominates: w_1 covers all ℓ_i and w_2; w_4, w_7, w_{10} cover w_3, w_5, w_6, w_8 and w_9, w_{11}; and w_{12} covers all ℓ'_j. The five closed neighbourhoods

$$\{\ell_1, w_1\}, \quad \{w_2, w_3, w_4\}, \quad \{w_5, w_6, w_7\}, \quad \{w_8, w_9, w_{10}\}, \quad \{w_{11}, w_{12}, \ell'_1, \dots, \ell'_b\}$$

of $\ell_1, w_3, w_6, w_9, w_{12}$ are pairwise disjoint, so $\gamma \geq 5$ by Lemma 3.1(2). Thus $\gamma = 5$.

By Lemma 4.2, $R(D(a,b)) = 2 + 1 \cdot 5 = 7 < 8 = \gamma_t(D(a,b))$. Finally, $12 + a + b$ takes every integer value $n \geq 22$ as (a, b) ranges over the pairs with $a, b \geq 2$ and $a + b \geq 10$. □

Remark 4.4. For $a + b = 10$ the degrees 1 and 2 both occur ten times, and WOWII definition 45 selects $\mathrm{mode}_{\min} = 1$. For $a + b \geq 11$ the mode is unique, so the refutation does not depend on the tie rule; the smallest such members have order 23. If $a+b = 9$, degree 2 becomes the unique mode, $R = 2+2\cdot 5 = 12$, and (1) holds for $D(a, b)$. The threshold is therefore sharp in the double-broom family.

5 One-ended brooms and the size of the failure

Definition 5.1. For integers $m \geq 3$ and $k \geq 1$, the *broom* $B(m, k)$ consists of a path $u_0 u_1 \cdots u_m$ and k further vertices $\ell_1, \dots, \ell_k$ adjacent only to u_0. It is a tree of order $m + 1 + k$.

Lemma 5.2. *If $m \geq 3$ and $k \geq \max\{2, m-2\}$, then $c(\langle N(L) \cup L\rangle) = 2$ and $\mathrm{mode}_{\min}(B(m,k)) = 1$; hence $R(B(m,k)) = 2 + \gamma(B(m,k))$.*

Proof. The leaves are $\ell_1, \dots, \ell_k, u_m$ and the supports are u_0 and u_{m-1}, which are not adjacent because $m \geq 3$. So $N(L) \cup L$ induces a star centred at u_0 and the edge $u_{m-1}u_m$, with no edge between them. The degree sequence has $k + 1$ ones, $m - 1$ twos and a single value $k + 1 \geq 3$; as $k + 1 \geq m - 1$, degree 1 is the smallest mode. □

Theorem 5.3. *For every $k \geq 10$, $\gamma_t(B(12,k)) = 8$, $\gamma(B(12,k)) = 5$ and $R(B(12,k)) = 7$. Hence there are also one-ended counterexamples of every order $n \geq 23$.*

Proof. The set $\{u_0, u_1, u_2, u_3, u_6, u_7, u_{10}, u_{11}\}$ is total dominating: the ℓ_i use u_0, and the path vertices $u_0, \dots, u_{12}$ use, respectively,

$$u_1,\ u_0,\ u_1,\ u_2,\ u_3,\ u_6,\ u_7,\ u_6,\ u_7,\ u_{10},\ u_{11},\ u_{10},\ u_{11}.$$

The vertices $\ell_1, u_0, u_3, u_4, u_7, u_8, u_{11}, u_{12}$ have the pairwise disjoint open neighbourhoods $\{u_0\}$, $\{u_1, \ell_1, \dots, \ell_k\}$, $\{u_2, u_4\}$, $\{u_3, u_5\}$, $\{u_6, u_8\}$, $\{u_7, u_9\}$, $\{u_{10}, u_{12}\}$, $\{u_{11}\}$. Hence $\gamma_t = 8$. The set $\{u_0, u_3, u_6, u_9, u_{12}\}$ dominates, and the closed neighbourhoods of $\ell_1, u_2, u_5, u_8, u_{11}$ are pairwise disjoint; hence $\gamma = 5$. Lemma 5.2 with $m = 12$ gives $R = 7$, and the order $13 + k$ takes every value $n \geq 23$. □

Lemma 5.4. *For $m \geq 3$ and $k \geq 1$,*

$$\gamma_t(B(m,k)) \geq \frac{m+2}{2} \qquad \text{and} \qquad \gamma(B(m,k)) \leq 1 + \left\lceil \frac{m-1}{3} \right\rceil \leq \frac{m+4}{3}.$$

Proof. Let $P = \{\ell_1, u_0, u_1, \dots, u_m\}$, of size $m+2$. The vertex u_0 has the two neighbours ℓ_1, u_1 in P; each u_i with $i \geq 1$ has at most the two neighbours u_{i-1}, u_{i+1}; and each ℓ_j has the single neighbour u_0. Lemma 3.1(3) gives the first bound. For the second, u_0 dominates all ℓ_j and u_1, and the path $u_2 \cdots u_m$ of order $m-1$ is dominated by $\lceil (m-1)/3 \rceil$ of its vertices: take $u_3, u_6, \dots$, adding u_m when $m \equiv 2 \pmod 3$. Finally $\lceil (m-1)/3 \rceil \leq (m+1)/3$. □

Theorem 5.5. *For $m \geq 15$ and $k \geq m-2$,*

$$\gamma_t(B(m,k)) - R(B(m,k)) \;\geq\; \frac{m-14}{6} \;>\; 0.$$

In particular the additive deficit in (1) *is unbounded.*

Proof. By Lemmas 5.2 and 5.4, $\gamma_t - R \geq \frac{m+2}{2} - 2 - \frac{m+4}{3} = \frac{m-14}{6}$. □

The next theorem places Conjecture 340 between a false and a true statement. Lemma 2.1 and Lemma 3.2 show that

$$\gamma_t(T) \leq c(\langle N(L) \cup L \rangle) + \alpha\, \gamma(T) \tag{2}$$

holds for every tree of order $n > 2$ when $\alpha = 2$, since $c(\langle N(L) \cup L \rangle) \geq 1$.

Theorem 5.6. *Let $\alpha < 3/2$ and C be real constants. There is a tree T of order $n > 2$ with* $\mathrm{mode}_{\min}(T) = 1$ *and*

$$\gamma_t(T) > c(\langle N(L) \cup L \rangle) + \alpha\, \gamma(T) + C.$$

Consequently, among trees whose smallest mode is one, inequality (2) *with an additive constant fails for every $\alpha < 3/2$ and holds for $\alpha = 2$.*

Proof. Take $T = B(m, m-2)$ with $m \geq 4$, so that $\mathrm{mode}_{\min}(T) = 1$ and $c(\langle N(L) \cup L \rangle) = 2$ by Lemma 5.2. If $\alpha \geq 0$, Lemma 5.4 gives

$$\gamma_t - 2 - \alpha\gamma - C \;\geq\; \frac{m+2}{2} - 2 - \alpha\,\frac{m+4}{3} - C = \Big(\frac{1}{2} - \frac{\alpha}{3}\Big)m - 1 - \frac{4\alpha}{3} - C,$$

which tends to infinity as $m \to \infty$ because $\alpha/3 < 1/2$. If $\alpha < 0$, then $-\alpha\gamma \geq 0$ and $\gamma_t - 2 - \alpha\gamma - C \geq \gamma_t - 2 - C \geq (m+2)/2 - 2 - C$, which also tends to infinity. In both cases a sufficiently large m gives the required tree. □

Question 5.7. What is the smallest constant $\alpha^* \in [3/2, 2]$ such that $\gamma_t(T) \leq c(\langle N(L) \cup L \rangle) + \alpha^* \gamma(T) + O(1)$ holds for all trees with $\mathrm{mode}_{\min}(T) = 1$?

6 Smallest counterexamples

Proposition 6.1 (computational)**.** *No tree of order at most* 21 *violates* (1)*. Exactly four trees of order* 22 *violate it, namely $D(2,8)$, $D(3,7)$, $D(4,6)$ and $D(5,5)$; each has $\gamma_t = 8$ and $R = 7$.*

This statement was obtained by exhaustive computation over all 9 114 283 unlabelled trees of orders 3 to 22, as described in Section 7. It is the only result of this paper that relies on computation. The following table summarises the census for the largest orders; the full per-order record is part of the supplement.

order n	trees	$\min_T (R(T) - \gamma_t(T))$	violations
19	317 955	0	0
20	823 065	0	0
21	2 144 505	0	0
22	5 623 756	-1	4
3–22	9 114 283		4

All orders $3 \le n \le 21$ have minimum slack 0, so the inequality is attained with equality at every small order but is never violated there.

7 Reproducibility

The supplement `wowii340-reproducibility.zip` requires only Python 3.10 or later and no third-party package. After extraction into an empty directory, the command

```
python -B -X utf8 reproduce340.py
```

verifies the SHA-256 hashes listed in `MANIFEST340.json`, copies the programs into a fresh temporary directory without any stored results, runs them there, and compares the complete regenerated JSON reports with the frozen reports in `expected/`. The default run replays the census up to order 18 in a few minutes. The option `--full` replays orders 3 to 22; on a single core of a current laptop this takes on the order of half an hour. The driver exits with a nonzero status on any mismatch, and prints `status: PASS` otherwise. Assertions must be enabled; both programs refuse optimised execution.

The certificate checker `verify340.py`. From explicit edge lists it recomputes γ_t and γ of $D(2,8), D(3,7), D(4,6), D(5,5)$, $B(12,10)$ and $B(12,11)$ by enumerating vertex subsets in increasing cardinality, without any dynamic program. It checks the total dominating sets, dominating sets and both packings of Theorems 4.3 and 5.3 for all 264 pairs $2 \le a \le b \le 24$ with $a + b \ge 10$ and for $10 \le k \le 40$. It also checks the counting and covering bounds of Lemma 5.4 and the inequality of Theorem 5.5 for $15 \le m \le 120$ and $k \in \{m-2, m-1\}$. Its controls confirm that a single vertex never totally dominates, that repeated edges are rejected, and that $D(4,5)$ and $B(12,9)$, just below the mode threshold, satisfy (1).

The census `census340.py`. Trees are generated by the constant-time free-tree successor algorithm of Wright, Richmond, Odlyzko and McKay [11], which lists each isomorphism class exactly once. The program checks completeness against the tree counts of OEIS A000055 [10] for every order. It checks distinctness by computing, for every tree, the Aho–Hopcroft–Ullman canonical code [1] of the tree rooted at its centre (the lexicographically smaller code at a bicentre), and requiring the 64-bit BLAKE2b digests of these codes to be pairwise different. The parameters γ_t and γ are computed by two separately written linear-time tree dynamic programs: one with per-vertex state arrays, the other by min-plus folding of child state vectors. On every tree of order at most 11 both are compared with exhaustive subset enumeration. For orders up to 16 the two programs are compared on every tree. For larger orders the second program is evaluated whenever the first gives $\gamma_t > c(\langle N(L) \cup L \rangle) + \mathrm{mode}_{\min}$, which is necessary for a violation because $\gamma \ge 1$; the programs must agree on all such trees. The census record lists, for each order, the number of trees, the minimum slack and every violation together with its parent array.

Scope of the checks. The two programs were written in the same research session from the same definitions; they are internal self-checks, not an external replication or a proof-assistant certificate. The universal statements, Theorems 4.3–5.6, are proved above and do not depend on them.

8 Source, status and priority

The statement and the linked definitions were taken from the Internet Archive capture of 23 July 2026 of the original WOWII pages [5]. The supplement retains the compressed capture bytes with their SHA-256 hashes, the extracted row 340, and the extracted definitions 45, 46, 49, 94 and 100, and the replay re-extracts them from the stored bytes. On 17 October 2026 the original endpoint redirected and then timed out. The archived row bears the status marker O. This is historical evidence of open status, not a statement by the curator about the present.

We found no earlier resolution. The 2009 paper of DeLaviña, Larson, Pepper and Waller [6] discusses other entries of the same block. A public worklog by Dubey [7] reports that entry 340 survives every tree of order at most 18, which agrees with Proposition 6.1. The archive of Bhattacharjee [2] refutes Conjectures 319, 352, 358, 359 and 427 and does not mention 340. The repository `formal-conjectures` [8] contains no formalisation of WOWII 340. A full-text scan on 17 October 2026 of all 244 issues and pull requests matching "WOWII", "WrittenOnTheWallII", "Written on the Wall" or "Graffiti.pc" found no mention of it; the repository was then at commit `c77fdd446d23`. Targeted web searches found nothing further. These bounded searches do not establish worldwide priority.

Acknowledgements and disclosure

The search for the construction, the proofs, the verification programs and the drafting were carried out with the assistance of Claude (Anthropic). The supplementary checks described in Section 7 are self-audits within that assisted workflow; no external peer review or proof-assistant certification is claimed. Jonas J. Gebendorfer is the sole author and takes responsibility for the content.

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